

Qikang Huang

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EDUCATION

University of Michigan, Ann Arbor

Sep. 2026 – May 2028

M.S. in Electrical and Computer Engineering

Track: Signal & Image Processing and Machine Learning

Communication University of China

Sep. 2022 – Jul. 2026

B.Eng. in Intelligent Science and Technology

- **GPA:** 88/100
- **Core Courses:** Large Models and Deep Learning (98), Neural Networks and Deep Learning (97), Natural Language Processing (89), Machine Learning (92), Data Structures (93), Computer Vision (91)

PROFESSIONAL EXPERIENCE

iFLYTEK Co., Ltd., Zhengzhou, China

Jan. 2026 – Apr. 2026

Assistant Research Algorithm Engineer — Smart Livestock Vision System

- Contributed to YOLOv8-based pig vision models for smart farming scenarios, including pig counting, posture detection, backfat-related classification, and abnormal-case monitoring.
- Refined the training-data feedback loop by validating 10,000+ RGB/depth images in LabelMe and correcting missed, false, duplicate, and inaccurate bounding-box annotations.
- Analyzed production failure cases from inspection robots and pig-counting modules, categorized false positives and false negatives, and supported model retraining and iteration.

Business Intelligence of Oriental Nations Corporation Ltd., Beijing, China

Jan. 2025 – Feb. 2025

Data Analyst Assistant — Telecom Traffic Revenue Forecasting

- Supported a telecom operator mobile data-traffic revenue forecasting project to predict future revenue and identify key business drivers.
- Analyzed traffic-revenue time series with ACF/PACF, built ARIMA, SARIMA, ETS, and Prophet models, and evaluated performance using MAE.
- Selected SARIMA as the best-performing model, completed a 14-month revenue forecast, and proposed optimization recommendations on traffic unit price, average usage, billed users, and zero-billed users.

RESEARCH EXPERIENCE

Retrieval Strategy Optimization for Multi-turn Legal Consultation Systems Based on RAG

Jan. 2026 – May 2026

Undergraduate Thesis

- Designed and implemented a RAG-based legal document retrieval optimization framework for multi-turn Chinese legal consultation, addressing context dependency, information omission, colloquial queries, and mismatch between user questions and formal legal expressions.
- Built an evaluation pipeline based on the LexRAG dataset, covering 17,228 Chinese legal articles and 5,065 multi-turn consultation queries, and systematically evaluated retrieval recall and ranking quality using Recall@K and NDCG@K.
- Integrated multiple retrieval and reranking modules, including query rewriting, BGE-M3 hybrid retrieval, HyDE query enhancement, and Cross-Encoder reranking, to improve candidate article recall and ranking performance.
- Experimental results showed that the HyDE-enhanced hybrid retrieval strategy achieved the best performance, improving Recall@10 from 0.3757 to 0.4701 and NDCG@10 from 0.2651 to 0.3349, providing empirical support for retrieval-chain optimization in legal RAG systems.
- Found that key bottlenecks in multi-turn legal consultation retrieval were more concentrated in query representation modeling and legal terminology expansion than in simple reranking of candidate results.

Loan Default Prediction Model Development and Optimization Based on Credit Data

Jul. 2025 – Sept. 2025

Alibaba Tianchi Financial Risk Prediction Competition

- Built a complete loan default prediction pipeline for credit risk control, covering data cleaning, feature engineering, model training, hyperparameter tuning, and performance evaluation to predict whether users would default on loans.
- Performed time-field decomposition, categorical feature encoding, numerical-variable binning, and construction of income-liability-credit correlation features, time-window statistics, and group-ratio features to enhance representation of repayment ability, debt pressure, and credit activity.
- Compared multiple tree-based models, including LightGBM, XGBoost, and CatBoost, and evaluated model performance using stratified K-fold cross-validation and ROC-AUC.
- Applied PSI-based feature stability selection, L2 regularization, and early stopping to reduce distribution shift and overfitting, and explored stacking fusion based on out-of-fold prediction results.
- Achieved a final optimized XGBoost model with an AUC of approximately 0.738, outperforming LightGBM and CatBoost in the experiment.

SKILLS & INTERESTS

- **Technical Skills:** Vibe Coding, Python, PyTorch, NumPy, OpenCV, Matplotlib, LaTeX
- **Languages:** English (IELTS: 7.0, GRE: 325), Chinese
- **Interests:** Fitness, Hiking